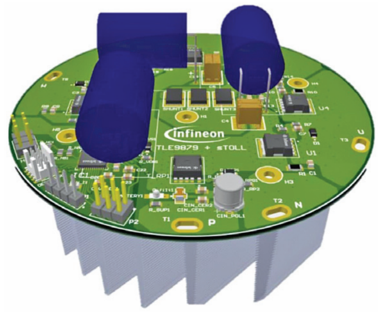
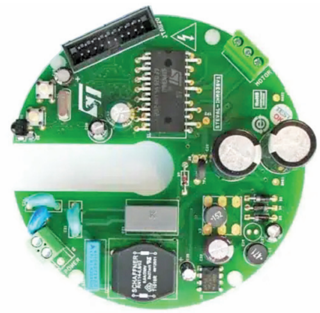
Interesting

Reference Designs OF FANS

Title of Reference Design	1kW Cooling Fan	BLDC Motor Driver Suitable for Fan Controllers
Image		
Key Attractions	The 1kW engine cooling fan reference design for 12V applications introduces a streamlined approach to automotive cooling systems, enhancing performance and compatibility for manufacturers.	The 3-phase BLDC motor drive reference design up to 50W for fan controllers is designed for ceiling fan applications. Features include efficient motor drive, power factor correction (PFC) and comprehensive hardware protection.
Highlights	Infineon Technologies has launched a 1kW engine cooling fan reference design for 12V applications, showcasing an automotive 3-phase motor drive. This design enables manufacturers to save time and resources in developing their cooling fan systems, ensuring high performance and compatibility. The MOSFETs used have a gate charge greater than 100nC, providing high current capability. The design features the world's first integrated bridge driver, capable of efficiently driving 1kW motors at 12V, setting a new standard in performance and integration for automotive cooling systems. The reference design includes an embedded power integrated circuit (IC) serving as a single-chip 3-phase motor driver, offering a comprehensive system-on-chip (SoC) solution. This device integrates an ARM Cortex M3 core, LIN transceiver, bridge driver, and power supply, enabling the implementation of advanced motor control algorithms like sensorless field-oriented control (FOC). Another key component is a 40V MOSFET offering a smaller form factor of 7×8 mm^2 while maintaining excellent thermal performance. The reference design also incorporates optimised thermal behaviour measures, a serial wire debug (SWD) port for easy debug connection, and dedicated ports for Hall sensors, LIN communication.	STMicroelectronics has launched the STEVAL-IHM038V1 reference design, a 3-phase BLDC motor drive up to 50W, suitable for fan controllers, utilising field oriented control (FOC) sensorless mode specifically designed for ceiling fan applications. The design, based on a 32-bit ARM Cortex M3 core microcontroller STM32F100C6T6B, integrates an inverter stage and control stage to drive a 3-phase BLDC fan motor in the power range of 30-35W. It incorporates a passive power factor correction (PFC) stage for up to 0.90 power factor, meeting minimum requirements. The design supports input voltages ranging from 90V AC or 128V DC to 265V AC or 375V DC, and the motor can handle a continuous output power of up to 50W. Hardware protection features include overcurrent and overtemperature protection using an NTC thermistor, as well as overvoltage and undervoltage detection mechanisms. The system is divided into four main blocks: control, power, motor, and power supply, with a PCB size tailored to fit a ceiling fan's form factor.
Applications	The reference design is suitable for an automotive 3-phase motor drive for engine cooling fan applications.	The reference design is specifically designed for ceiling fan applications.
OEM Brand	Infineon Technologies	STMicroelectronics
URL	https://www.electronicsforu.com/electronics-projects/reference-design-for-1-kw-cooling-fan	https://www.electronicsforu.com/electronics-projects/industry-grade-projects/reference-design-for-bldc-motor-driver-suitable-for-fan-controllers